# Oluwatosin Ayeni

ML Runtime Engineer — Fractile

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## PROFILE

ML Runtime Engineer with deep hands-on experience in vLLM and SGLang runtime integration, KV cache kernel profiling, and paged attention characterisation. Benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL — 1,443× TTFT speedup, 5,965 tok/s peak, 100% SLO compliance. Profiled CUDA KV cache ops on RTX 4090 with NVIDIA Nsight: identified memory bandwidth bottleneck, 3.8× throughput improvement, 18× latency reduction. Open to London or Bristol relocation.

## CORE COMPETENCIES

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|---|---|
| **Inference Runtimes** | vLLM, SGLang, TensorRT-LLM, ONNX Runtime — integration, profiling, KV cache management, paged attention |
| **GPU & CUDA** | NVIDIA Nsight Systems/Compute, CUDA kernel profiling, memory bandwidth analysis, warp occupancy, kernel access patterns, BFloat16/FP8 |
| **KV Cache & Attention** | Paged attention scheduling, KV block eviction, prefix/RadixAttention caching, memory pressure characterisation, flash attention |
| **Inference Research** | Speculative decoding, continuous batching, AWQ/GPTQ quantisation, tensor parallelism, GQA/MQA/MLA architectures |
| **Infrastructure** | Kubernetes (HPA, GPU pools), Docker, Prometheus, Grafana, FastAPI, Python |

## PROFESSIONAL EXPERIENCE

Jan 2025–present
**Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present
**Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present
**Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present
**AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

## KEY PROJECTS

**llm-inference-benchmark (H100)** vLLM v0.30 vs SGLang v0.5.20, 2× H100 NVL 94 GB, Qwen3-8B BF16, TP=2. Characterised paged attention and KV cache scheduling under memory pressure at 131K-token context. RadixAttention: 1,443× TTFT speedup. Live results

**open-model-engine** GPU inference benchmarking: CUDA profiling, AWQ/GPTQ, speculative decoding experiments. GitHub

**phase2-llm-inference-deep-dive** CUDA KV cache kernel profiling: memory bandwidth bottleneck identification, 3.8× throughput improvement, 18× latency reduction. GitHub

**phase3-model-architecture** Transformer internals, GQA/MQA/MLA attention variants, VRAM modelling, speculative decoding experiments. GitHub

**vllm-k8s-stack** Production Kubernetes LLM serving: KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana. GitHub

## EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

## OPEN SOURCE

**GitHub** 43 public repos — github.com/AyeniOluwatosinOlawale